Gauge saturation, orbifold compactification, and marked smoothing realization

24 September 2026. A continuation of the actual-form composition and compact KE realization. This note removes the assumption that the second homotopy group vanishes from the **compactification** conclusion. Smooth realization is then obtained under a separate, explicit algebraic and relative-diffeomorphism condition. It does not assert that these latter conditions follow from Ricci pinching.

The contribution here is a composition of the inherited actual-form estimates with classical bounded-Ricci orbifold compactness and algebraic smoothing. The compactness, ALE classifications, deformation criterion and negative KE existence theorem retain their classical attributions. No priority claim is made for the composition.

1. Original input and compactification theorem

All manifolds are connected, closed, smooth and oriented of real dimension four. Ricci uses the tensor norm except in the definition of the pointwise operator defect. Two-forms use exterior-algebra norms and Weyl curvature uses the Hilbert–Schmidt norm as an endomorphism of two-forms. Write

$$\delta_j = \|\operatorname{Ric}^{\sharp}_{g_j} + I\|_{\infty,\mathrm{op}}, \quad V_j = \operatorname{Vol}(M_j, g_j), \quad p_j = (a_j^+)^2,$$

$$q_j = \frac{\mathfrak{D}_2(g_j, a_j)}{V_j}, \qquad \mathfrak{D}_2(g, a) = \frac{3}{2}\int_M (s^2 + 12|W^+|^2) - 72\pi^2 (a^+)^2. \tag{1}$$

An **actual conformally solvable class** means the Chern class of a Spin^c structure whose unperturbed monopole equations are solvable for every metric in the conformal class of the specified metric. The structure may vary with j, but is fixed when its auxiliary positive weights vary.

Theorem A (negative KE orbifold compactification). Suppose that, for fixed $D, A_0 > 0$,

$$\operatorname{diam}(M_j, g_j) \leqslant D, \qquad p_j \geqslant A_0, \qquad \delta_j + q_j \longrightarrow 0, \tag{2}$$

and each a_j is an actual conformally solvable class. No hypothesis is made on $\pi_1(M_j)$ or $\pi_2(M_j)$. A subsequence converges in the Gromov–Hausdorff topology to a compact oriented orbifold (X, h) with finitely many isolated quotient singularities, and

$$\operatorname{Ric}_h = -h.$$

There is a **global orbifold Kähler structure** J on X, with its given orientation. On every compact subset of X_{reg}, the convergence is realized by smooth embeddings and is $C^{1,\alpha}$ for every $\alpha < 1$ and strong $W^{2,p}$ for every finite p. The limiting metric and complex structure are smooth in orbifold charts.

No covering space is required to define J. No negative-square cost, Ricci-derivative bound, curvature-energy assumption, or initially given complex structure on M_j is added to (2). This is not smooth compactness on all of M_j: topology may be carried by the neighborhoods that shrink to the singular points.

Proof: the bounded-Ricci compactness input really applies

Put $r_j = \fint |\operatorname{Ric}_{g_j} + g_j|^2$ and $\kappa_j = r_j + q_j$. For large j, $\delta_j \leqslant 1/2$, so

$$|\operatorname{Ric}_{g_j}| \leqslant 3, \qquad -6 \leqslant s_j \leqslant -2, \qquad r_j \leqslant 4\delta_j^2.$$

The actual monopole scalar inequality gives

$$32\pi^2 A_0 \leqslant \int s_{j,-}^2 \leqslant 16(1+\delta_j)^2 V_j, \qquad V_j \geqslant \frac{8\pi^2 A_0}{9} =: v_0. \tag{3}$$

Bishop–Gromov comparison using the diameter bound provides an upper volume bound $V_j \leqslant V_0(D)$ and, at every point, lower ball-volume bounds $\operatorname{Vol} B_\rho \geqslant c(D, A_0)\rho^4$ for $0 < \rho \leqslant 1$. Applying the local Jiang–Naber L^2-curvature bound on a uniformly finite fixed-scale cover yields

$$\int_{M_j} |\operatorname{Rm}|^2 \leqslant E(D, A_0). \tag{4}$$

Use **bounded-Ricci orbifold compactness**, not an Einstein-only bubble theorem: two-sided Ricci bounds, noncollapse, bounded diameter and bounded critical curvature energy give an orbifold subsequential limit with $C^{1,\alpha}$, weak $W^{2,p}$ orbifold metric and regular-locus convergence. Anderson explicitly distinguishes this result from the earlier Einstein theorem in the introduction of [A04], referring to [A90]. The varying-manifold version uses (4). The same bounded-Ricci conclusion is used explicitly in Honda–Mondino [HM25], proof of Theorem 2, p. 452. The Ricci lower bound gives ordinary, irreducible quotient neighborhoods, not a union of several cones at one vertex; see [A04], Remark 2.1(i).

On regular harmonic charts the metric equation is

$$g_j^{ab}\partial_a\partial_b(g_j)_{uv} = -2(\operatorname{Ric}_{g_j})_{uv} + Q_{uv}(g_j, \partial g_j). \tag{5}$$

The Ricci error itself tends uniformly to zero. The limiting equation is $\operatorname{Ric}_h = -h$; elliptic bootstrapping makes h smooth on the regular locus. Applying interior estimates to the difference of (5) and its limiting equation gives strong $W^{2,p}$ convergence on smaller charts. The right-hand side converges strongly; no derivative of the Ricci error is needed. The limit already has $W^{2,p}$ regularity in local orbifold covers by the compactness theorem. Its Einstein equation holds almost everywhere there (a point has zero measure); harmonic-coordinate regularity and elliptic bootstrapping therefore make h a smooth Einstein orbifold metric at the singular points as well.

Proof: saturation selects a global complex structure

The inherited actual-form theorem supplies smooth self-dual ω_j and choices $\eta_j \downarrow 0$ with

$$\|\omega_j\|_\infty \leqslant 2\sqrt{3}, \qquad \fint (|\omega_j|^2 - 2)^2 \leqslant 720\kappa_j + \eta_j,$$

$$\fint |\nabla\omega_j|^2 \leqslant 13\kappa_j + 4\sqrt{2\kappa_j} + \eta_j. \tag{6}$$

Only the first-order part of that theorem is used. In particular its additional second-order estimate and square-cost hypothesis are not used. Because $V_j \leqslant V_0$, the unnormalized derivative and amplitude errors in (6) tend to zero. Pull back to the same regular-locus embeddings as the metrics.

Rellich compactness on an exhaustion and a diagonal subsequence give a global weak limit ω_∞. It satisfies

$$\nabla^h \omega_\infty = 0, \qquad *_h \omega_\infty = \omega_\infty, \qquad |\omega_\infty|^2 = 2. \tag{7}$$

The norm assertion follows from strong local L^2 convergence, the uniform amplitude bound and (6); it is not inferred merely from nonzero total mass. The regular locus is connected. The form is one global limit of global forms, rather than locally chosen eigenvectors.

In a uniformizing ball at a singular point, lift (7) to the punctured ball. The smooth extended metric has bounded connection coefficients. The parallel equation, constant norm and short paths imply that the lift has a unique continuous limit at the origin. Its first-order parallel equation then gives smooth extension. The extension is invariant under the finite chart group because the form was invariant on the punctured ball. Thus (7) holds on the whole orbifold. The relation $\omega_\infty(u,v) = h(Ju,v)$ defines a parallel orthogonal complex structure: a self-dual two-form of norm $\sqrt{2}$ in dimension four squares to $-I$ in this relation. Parallelism gives integrability and closedness. This proves Theorem A. □

2. What the small-scale fillings do, and do not, inherit

There is a useful conclusion stronger than an unconstrained Ricci-flat blow-up. Set $w_j = \sqrt{6}|W^+_{g_j}|$. The classical completed-square estimate used in the actual-form proof gives

$$⨍ (w_j - 2)^2 \leqslant 2\kappa_j, \qquad \int_E |W^+_{g_j}|^2 \leqslant \frac{4}{3}\,\mathrm{Vol}(E) + \frac{2}{3}\kappa_j V_j \tag{8}$$

for every measurable subset $E \subset M_j$.

Proposition B (chirality of smooth ALE bubbles). Any smooth complete nonflat pointed limit obtained by harmonic-radius blow-up at scales $\rho_j \to 0$, with convergence on compact subsets, is Ricci-flat and anti-self-dual in the inherited orientation. With the Euclidean growth and finite energy supplied by (3)–(4), it is ALE.

Indeed $|\mathrm{Ric}_{\rho_j^{-2} g_j}| \leqslant 3\rho_j^2$. On each rescaled ball of fixed radius R, (8), scale invariance of four-dimensional Weyl energy and the unscaled upper ball-volume bound give

$$\int_{B_R} |W^+_{\rho_j^{-2} g_j}|^2 \leqslant C\rho_j^4 R^4 + \frac{2}{3}\kappa_j V_0 \longrightarrow 0.$$

Harmonic-coordinate convergence as in (5) is strong enough to pass curvature in L^2. The limit has $W^+ = 0$. Bando–Kasue–Nakajima's asymptotic theorem applies to this **exact smooth Ricci-flat limit**, not to the input metrics. This proves the proposition.

Wright's classification [W11], Theorem A, now applies legitimately: a non-simply-connected nonflat bubble is a finite cyclic quotient of a Gibbons–Hawking space and is Kähler; simply connected bubbles are the hyperkähler ALE spaces. Şuvaina's [S11], Theorem B, relates the resulting Ricci-flat Kähler complex surface to a **minimal resolution of a fiber** of a local Q-Gorenstein deformation. The theorem does not say that the particular complex structure of every bubble is already a smooth Milnor fiber. Nor has the chosen bubble complex structure been identified with the limiting complex marking at the outer orbifold link.

In particular, Proposition B is not a classification of the entire compact region removed from M_j at a singular point. Such a region can involve several scales and intermediate orbifold levels. No

bubble-tree energy identity or assertion that its topology equals one terminal bubble is used below. The realization condition is imposed on the **whole actual filling**.

A concrete distinction is the quotient singularity $\mathbb{C}^2/\frac{1}{4}(1,1)$. Its minimal resolution has a single (-4)-sphere, so a compact resolution neighborhood has $b_2 = 1$. Its Q-Gorenstein Milnor fiber is instead a rational ball with $b_2 = 0$ and $\pi_1 = \mathbb{Z}_2$. They fill the same oriented quotient link but are not diffeomorphic. This classical example already prevents identifying a smoothing filling from the link alone. It is not claimed that both fillings arise from sequences satisfying (2).

3. A marked smoothing criterion returning to the original smooth type

Let (X, h, J) be the limit in Theorem A, with singular points $x_1, \dots, x_N$. Choose small disjoint quotient neighborhoods U_ν with smooth links Y_ν, and put

$$K = X \setminus \bigcup_\nu \operatorname{int} U_\nu.$$

Regular-locus convergence gives oriented smooth embeddings $\iota_j : K \hookrightarrow M_j$, including collars of the boundary. Let B_j be the closure of their complement. The condition below includes that B_j has exactly one connected filling $B_{\nu,j}$ at each boundary component, so no unproved topological separation statement is hidden in the notation.

A **projective Q-Gorenstein smoothing** is a proper flat projective family $\pi : \mathscr{X} \to \Delta$ with central fiber X, smooth fibers X_t for sufficiently small $t \neq 0$, and relative canonical divisor $K_{\mathscr{X}/\Delta}$ Q-Cartier. Use the complex structure J just produced, not an unrelated complex structure on the underlying space. After shrinking Δ, the exterior is differentiably trivial by Ehresmann's theorem with boundary. Its identification with K equips each local Milnor fiber $F_{\nu,t}$ with a specified boundary marking by Y_ν.

Theorem C (marked Q-Gorenstein realization). Suppose X admits such a smoothing. For an index j, assume that for some small regular parameter $t_j \neq 0$ each whole filling admits an orientation-preserving diffeomorphism

$$f_{\nu,j} : B_{\nu,j} \longrightarrow F_{\nu,t_j} \tag{9}$$

whose boundary restriction agrees, up to isotopy, with the marking induced by ι_j and the exterior trivialization. Then the same oriented smooth manifold M_j admits a negative KE metric. If (9) holds for all sufficiently large indices of the extracted subsequence, the conclusion holds for all of those indices.

Proof. An isotopy on a boundary extends across its collar. Modify the maps (9) there so that they agree exactly with the exterior identification. Collar-compatible gluing then gives an oriented diffeomorphism

$$M_j \cong X_{t_j}. \tag{10}$$

The KE orbifold has positive canonical orbifold bundle: the curvature form of its canonical Hermitian metric is $-\rho_h = \omega_h > 0$. A divisible power of the canonical bundle descends to the quotient space, and the orbifold Kodaira embedding theorem makes K_X ample as a Q-divisor. Equivalently this is the usual canonical polarization of a negative KE orbifold with quotient singularities.

Choose an index m such that $mK_{\mathscr{X}/\Delta}$ is Cartier. Its restriction to X is ample. Openness of relative ampleness, after shrinking the base, makes K_{X_t} ample on each nearby smooth fiber. The Aubin–Yau

theorem gives a smooth Kähler metric h_t with $\mathrm{Ric}_{h_t} = -h_t$ on X_t. Pulling it back through (10) proves the claim. □

Thus the output meets the Yau26 existence requirement on the covered smooth types. It need not be close to g_j, and no invertibility or integrability assumption on a tensor Einstein gluing operator is needed. The analytic compactification and the marked smooth-type matching remain different steps.

The relative condition (9) is finite local topology plus gluing data. It is not replaceable by equality of fundamental groups, Betti numbers, links or intersection forms. It is also not inferred from the ALE classification. Once the marked data are known, however, no global diffeomorphism or complex structure on M_j has to be assumed: (10) constructs the former and pulls back the latter.

4. An algebraically checkable smoothing input

For quotient surface singularities, the local Q-Gorenstein smoothable types are the rational double points and the cyclic singularities

$$\frac{1}{dn^2}(1, dna - 1), \qquad d > 0, \quad n \geqslant 2, \quad (a, n) = 1. \tag{11}$$

These are the class T singularities of Kollár–Shepherd-Barron. A local smoothing at each singularity does not itself supply a global smoothing.

Corollary D (cohomological realization test). In Theorem C the global smoothing assumption is satisfied if the normal projective surface X has only class T singularities and its minimal resolution $r : Z \to X$, with reduced exceptional divisor E, satisfies

$$H^2(Z, T_Z(-\log E)) = 0. \tag{12}$$

Consequently (11), (12) and the marked filling condition (9) imply negative KE existence on the original M_j.

This is exactly the globalization criterion of Lee–Park [LP07], Theorem 2.1; their proof passes from (12) to vanishing of the global obstruction sheaf cohomology, then globalizes the local Q-Gorenstein deformations. We use its projective family, not an arbitrary first-order deformation. Condition (12) is sufficient, not necessary, and is not claimed to follow from the Ricci or gauge estimates.

For rational double points the standard Milnor filling is diffeomorphic to the ADE resolution plumbing. For the cyclic types the Milnor filling, not generally the full exceptional plumbing, is the relevant object; the case $d = 1$ is the usual rational-ball replacement. Răsdeaconu–Şuvaina [RS09], Proposition 2.6 and Remark 2.9, give the classical algebraic rational-blowdown interpretation. To apply this interpretation to an arbitrary M_j, its actual exterior and boundary marking must still match the algebraic surgery. Theorem C expresses precisely that requirement.

Concrete ADE specialization. Suppose the limit complex surface is a degree $d \geqslant 5$ hypersurface in $\mathbb{CP}^3$ with only nodes. A generic pencil $F_0 + tG = 0$, after restricting to a small disk, is a global Gorenstein smoothing; adjunction gives $K_{X_t} = \mathscr{O}_{X_t}(d - 4)$, which is ample. Each local Milnor fiber has the oriented smooth type of the (-2)-disk bundle over S^2. If the actual complementary pieces are these bundles with the Milnor boundary markings, Theorem C applies. This specialization needs no verification of (12), because the global smoothing is written explicitly. It is a classical family illustrating a nonempty, checkable matching test, not a newly constructed family of Einstein manifolds. In particular a vanishing π_2 condition is absent, not reintroduced under another name.

5. Precise scope and the remaining implication

The conclusions established from the raw data alone are

$$\boxed{(2) \implies \text{negative KE orbifold limit}}$$

and the anti-self-duality of its smooth Ricci-flat ALE blow-ups. With the additional global smoothing and marked filling inputs, they become

$$\boxed{\begin{array}{c}\text{negative KE orbifold} + \text{global QG smoothing} \\ +\text{marked Milnor fillings} \\ \Downarrow \\ \text{negative KE on the original smooth type.}\end{array}}$$

These statements handle a genuine alternative to excluding all bubbles: allow singular limits and recover the smooth type by algebraic realization.

There is no proof here that (2) forces all singularities to be class T in their limiting complex representation, that their local smoothings globalize, or that each entire geometric filling has the correct relative Milnor type. Classifying terminal ALE spaces does not prove those three claims. Resolving them, or supplying a counterexample to one, is the actual remaining issue for unconditional realization without the topological no-bubble hypothesis. Neither a full energy identity nor a new numerical gap constant substitutes for this missing information.

Primary references and exact roles

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Theorems A and C use only the parts of these results specified in their proofs. Baily and Aubin–Yau are invoked as classical existence/embedding theorems, not reproved here. The first-line actual-form estimates remain an inherited proof dependency, not an external literature theorem.